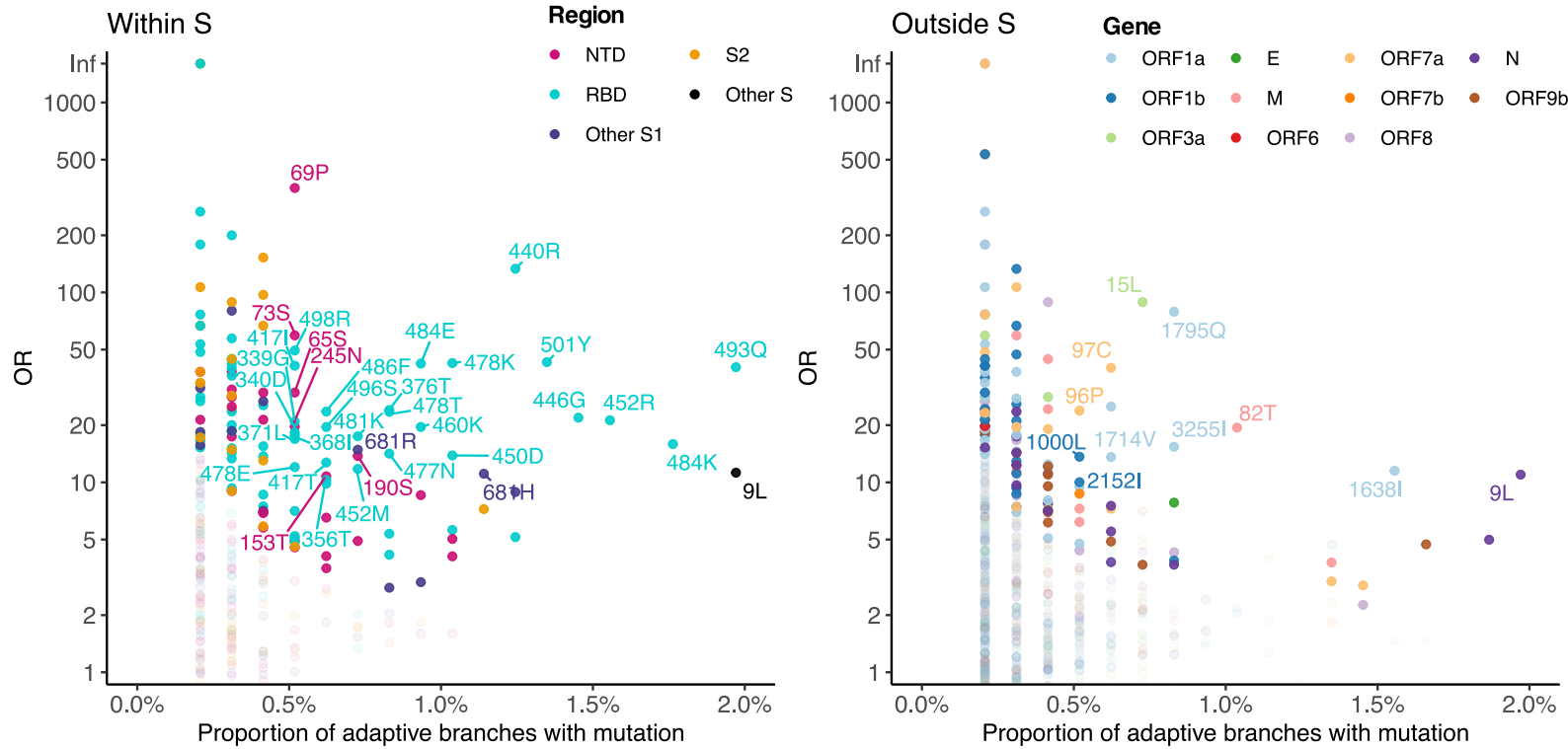

Within S
Region
NTD
RBD
Other S1
S2
Other S
OR
Inf
1000
500
200
100
50
20
10
5
2
1
69P
440R
73S
498R
417I
65S
339G
245N
340D
486F
484E
478K
501Y
493Q
496S
376T
478T
446G
452R
481K
460K
371L
368I
681R
484K
450D
478E
417T
477N
681H
9L
190S
452M
153T
356T
0.0%
0.5%
1.0%
1.5%
2.0%
Proportion of adaptive branches with mutation
Outside S
Gene
ORF1a
ORF1b
ORF3a
E
M
ORF6
ORF7a
ORF7b
ORF8
N
ORF9b
15L
1795Q
97C
96P
82T
1000L
1714V
3255I
2152I
1638I
9L